

Faculty of Natural and
Mathematical Sciences
Department of
Engineering/Information

King's College London
Strand Campus, London,
United Kingdom



7CCSMPRJ/7CCSMUIP

Individual Project Submission 2022/23

Name: Rupali Banerjee
Student Number: k21172849
Degree Programme: MSc Advanced Software Engineering
Project Title: Improve student's industry readiness through Project Based Learning application
Supervisor: Leonardo Magela
Word Count: Word count goes here

RELEASE OF PROJECT

Following the submission of your project, the Department would like to make it publicly available via the library electronic resources. You will retain copyright of the project.

Check the appropriate box below

- ☐ I **agree** to the release of my project
☐ I **do not** agree to the release of my project

Signature: *Signature*

Date: July 13, 2023



Department of Engineering/Information
King's College London
United Kingdom

7CCSMPRJ/7CCSMUIP Individual Project

Improve student's industry readiness through Project Based Learning application

Name: **Rupali Banerjee**
Student Number: **k21172849**
Course: **MSc Advanced Software Engineering**

Supervisor: Leonardo Magela

This dissertation is submitted for the degree of MSc in **MSc Advanced Software Engineering**.

Abstract

This Dissertation project aims to improve graduate employability by enhancing the learning techniques followed by students. The learning technique discussed extensively throughout the report is active learning that involves student participation in the learning process. Project Based Learning (PBL) is the approach used to aid active learning in STEM curriculum. The web application will encourage students to participate in PBL approach by providing them an opportunity to work on real world projects offered by organizations. This will not only improve their resume skillset but also provide them with the necessary confidence that needed to ensure they have a smooth transition at their workplace. Students will initially have the flexibility to enroll in assignments of their choice. Later on based on the score they received for the completed assignments, lecturers will assign them projects published by mentors (external stakeholders from different organizations). Students will be working on subtasks of specific projects, multiple students can work on different subtask of the project. This will give them a hands-on experience of agile development process that's followed by most organizations

Contents

1	Introduction	1
1.1	Aims and Objectives	1
1.2	Motivation	1
2	Literature Review	2
2.1	PBL in Software Engineering	3
2.2	PBL in Business and Entrepreneurship	5
2.3	PBL for Professional Training	6
2.4	Conclusion	7
3	Conceptualization	9
3.1	Deliverable	9
3.2	Platform	9
3.3	Tools and Technologies	9
3.3.1	Frontend	10
3.3.1.1	React.js	10
3.3.1.2	Material UI	10
3.3.1.3	Redux Toolkit	11
3.3.1.4	Npm Packages	11
3.3.2	Backend	11
3.3.2.1	Node.js	11
3.3.2.2	Express.js	11
3.3.3	Database	11
3.3.3.1	MongoDB	11
3.3.3.2	Mongoose	12
4	Application Design Overview	13
4.1	Feature and Functionality	13
4.2	Software Architecture	13
5	Implementation	14
5.0.1	Application flow	14
5.0.2	Unit testing	14
6	Conclusion	14
7	Future Scope	14
	References	15
A	Appendix	16
A.1	Points to Note	16

B	Review of stochastic calculus	17
B.1	Riemann integration	17
B.2	The Itô integral	17

List of Figures

1	Research Results.	5
2	Research method.	6

List of Tables

1 Introduction

Classroom-driven education encourages learners to actively engage in lectures or presentations, effectively grasp the content, and apply these concepts through homework, and assignments. However, it is worth considering that these educational programs might play a key role in students' ability to apply their knowledge to practical circumstances or solve problems on their own.[1]

Project-based learning (PBL) serves as an instructional framework that encourages participants through active analysis while honing essential problem-solving competencies. Thanks to its impact on student skill growth rates, PBL has reclaimed significance across academic environments globally. The benefits highlighted by research regarding project-based learning applications are increased engagement, better outcomes, and improved collaboration for students. This shows how effective these teaching tools can be in the academic environment.[1][2]

Apart from Project-based learning here are few other active learning techniques like:

1. **Gamification:** By including educational games, tests, or interactive exercises, the learning process can be gamified to increase participation and make the process entertaining.
2. **Concept Mapping:** Offer students the task to create concept maps for specific domain-based projects that improve critical thinking, and aid students in meaningfully organizing their knowledge.
3. **Socratic Seminars:** Conduct Socratic seminars where students engage in open-ended discussions on specific STEM topics. Through insightful questions, the instructor facilitates the discussion. This method encourages critical thinking, effective communication, and a greater understanding of STEM subjects.

1.1 Aims and Objectives

The objective of this project is to make learning an interesting and interactive process that enhances student's eagerness to learn and apply the concepts taught in the curriculum. This will help students gain better hands-on experience and improve their employability. The benefits and drawbacks of the current learning process and its enhancements are discussed in detail throughout the report. The project focuses on developing an application for students to apply computer science (CS) concepts discussed during classroom-based or blended learning approaches, to solve real-world problems through small-scale projects and eventually get an opportunity to work on projects offered by organizations.section 2.

1.2 Motivation

In classroom-based learning, students are always expected to just passively listen and make notes whenever required, whereas Project-based learning actively engages students

and requires them to think critically and creatively. The current formal education system curriculum has standardized assessment techniques to measure student achievement, whereas project-based learning uses a variety of assessment methods that reflect real-world skills and knowledge.

According to research, learners have trouble solving problems because they lack the appropriate knowledge and skill sets. Learning happens only when students have understood the underlying concepts and framework and are in a position to apply them in a real-world scenario, which ultimately brings us to the fact that application-based learning is most important if we have to improve the efficiency of the learning process and get the knowledge embedded

Currently being a master's student with almost four years of industry experience has made me realize the importance of problem-solving skills for efficient industry readiness. When I started my professional journey there was a huge gap in the curriculum and real-world domain-specific tasks, which can at times feel quite overwhelming as a fresher

The main objective of this project is to improve the student learning cycle through the inclusion of interactive learning methodology in the curriculum.

Educators employ the steps below to improve the efficiency of the learning process:

1. Professor uses different approaches to explain new concepts to students and assign them tasks (**Engage**)
2. Students **Explore** concepts with new materials provided by the professor or take up new assignments or experimental activities.
3. The students take up projects and start implementing them, posing questions on forums to clarify their doubts. The professors/mentors provide the details, respond to student queries (**Explain**)
4. The real-world projects challenge students' conceptual knowledge and technical proficiency. (**Elaborate**)
5. The teacher/industry expert **Evaluates** student learning outcomes throughout and at the end of the project completion and provides valuable feedback for improvement.

The above 5E instructional model is from the 5E instructional Model[3]

2 Literature Review

In project-based learning (PBL), students work on assignments that resemble those found in organizations with an emphasis on active engagement and critical thinking. In recent years, the use of project-based learning applications has grown in popularity to support project-based learning in STEM education. PBL facilitates students to identify their learning requirements, apply knowledge and skills to solve challenges and combine theory and practical application.[4][5]

In PBL, students work on tasks that are designed by professors, allowing them to cover important pre-planned educational material. Each project eventually results in a product made by the students that serve a purpose. The emphasis is on delivering excellent work that addresses the central query and is valuable to students and relevant beyond the university curriculum[4]. However, this approach of active learning cannot be used for modules like arts, psychology, or improving language fluency. Project work mainly involves the creation of UML diagrams for data flow, and control flow, incorporating feedback and suggestions provided by professors and other stakeholders. These projects also allow flexibility for student creativity and direction in creating output.[6]

Instead of relying on memorization, this technique(PBL) anticipates that students will develop and interpret the recently acquired knowledge (Kızıkan & Bektaş, 2017). Chiu (2020) argues in favor of the proposition that PBL can improve higher-order thinking abilities as opposed to data memorization. According to (Tseng et al. (2013)) PBL offers opportunities for meaningful learning and enables students to expand their knowledge through exploration.[7]

The impact of PBL and educators' direct instruction on students' academic competency in post-secondary education was assessed in Chen & Yang's 2019 review. According to this study, PBL refers to a learning process in which students are actively involved in resolving problems encountered during the implementation of projects. The findings indicated that PBL, as opposed to direct instruction, had a more favourable impact on student's academic progress.[8]

Most future-oriented professions require employees to be innovative and apply the principles and practices of Software Engineering they have learned in university. However, it is observed that students graduating with fundamental computer science, and software engineering literacy don't have a clear idea about how these concepts can be applied in practice. These studies indicate that when students engage in PBL, they are exposed to enterprise-level circumstances that are similar to those they faced when working on small projects, which results in increased professional involvement and a perception of clarity.[3]

The book "STEM Project-Based Learning: An Integrated Science, Technology, Engineering, and Mathematics (STEM) Approach. "highlights the interdependence between learning and career success as one of the most fundamental principles of PBL in STEM. Students should be provided with opportunities to work on interdisciplinary projects that require them to integrate their knowledge and skills from across all STEM subjects. It suggests that the knowledge and expertise students get through PBL activities are crucial for success in both academics and future careers.[2]

2.1 PBL in Software Engineering

Over the years the software engineering curriculum has been updated to ensure that formal education supports the acquisition of different skills such as problem-solving, teamwork, leadership, initiative, creativity, analytical thinking, and a few other soft skills. It is critical that learning software engineering or computer science as an academic discipline provides robust ties between theory and practical application. The

National Association of Colleges and Employers publishes a report each year to identify the qualities that hiring managers prefer to see in student resumes. According to its 2019 report [14], the three qualities (out of 20) deemed most crucial are communication skills, problem-solving abilities, and the ability to work in a team.[9]

According to the case study [13], PBL aided students in improving their average overall score by 18.26%. Most of the curriculum for the Software Engineering (SE) course, which was the core subject in this study, includes topics like human-computer interaction (HCI), generic object-oriented analysis and design challenges, software development processes, and UML. Large software systems analysis and design make use of these concepts and principles. The PBL learning technique has been demonstrated to have a significant impact as seen by the 1.9% rise in the overall average of PBL students. Student feedback for the above PBL case study was conducted to reflect upon their experience and suggestions regarding the overall process for the academic year (2018/19). 82% claimed that they were satisfied with the approach, 63% claimed that they have enjoyed the experience and 19% are neutral.[9]

There have been studies and experiments conducted in the past that allowed students to undertake role play in a software project setting that is close to a real-life opportunity. Students would be divided into groups, and everyone would take on roles aligned with Architect, Developer, Business Analyst, Tester, Configuration and Release engineer, etc with a typical team size of seven. There would also be an external stakeholder who performs the role of a business sponsor or Product Owner and might be from a commercial or government background. As the experiment progressed, students were encouraged to share feedback. A large majority of students gave positive feedback stating that they enjoyed such learning methods as it enhances their abilities to apply knowledge along the way. However, there have also been instances when neutral or negative feedback has been noticed. Such experiences have been linked to the self-directed learning requirements of PBL.[6]

Studies have established that partnering students with external stakeholders has the potential to improve overall outcomes. There are two key ingredients that could lead to successful outcomes. Firstly, students themselves need to be (self) motivated to undertake learning that is application-oriented; and secondly, in case they face challenges, the external stakeholders provide appropriate guidance and direction to overcome hurdles. Such collaboration allows a mutual learning experience that is rewarding for both students and stakeholders. In many ways it will also allow the curriculum to be fine-tuned based on students' feedback and learning experiences, so it is most relevant and useful.[6]

Numerous studies indicate that the implementation of PBL can amplify academic accomplishments, encourage participation, and promote confidence in one's scholarly aptitude. Employers today are searching for Software Engineers with technical and behavioural abilities, such as creativity, critical thinking, leadership, and communication, due to the fierce competition and quick technological advancements in the software sector. Employers assess some of these skills through reports of academic assignments that called for teamwork and critical thinking. To prepare students for a successful career in software engineering (SE), they must acquire training in programming and

requirements engineering (RE), emphasized in the ACM / IEEE curriculum guidelines and standardized in ISO / IEC / IEEE 12207: 2017 Systems . This indicates that the inclusion of PBL in academic training is crucial to improve graduate employability [1]

2.2 PBL in Business and Entrepreneurship

To evaluate the benefits of PBL, a case study was conducted at the School of Information Technology (IT) at Mae Fah Luang University (MFU) in Chiang Rai, Thailand, with 42 Thai undergraduates majoring in Digital Technology for Business Innovation (DTBI). From August 2020 to December 2020, they were enrolled in the course "Digital Business Initiation in Practises." [4]

The research methodology depicted in Fig. 1 has been used for the case study. All the enrolled students were segregated into 11 groups based on their selection, with 3-4 members per group. The results after implementing the methods are mentioned in Fig.2 The research results displayed a considerable improvement from the first presentation in the percentage of the student group from the second presentation for each viewpoint on the evaluation criteria.

Evaluation criteria	1st Presentation		2nd Presentation	
	<i>Number of qualified groups</i>	<i>%</i>	<i>Number of qualified groups</i>	<i>%</i>
Students can apply their knowledge for analysing and solving problems	5	45.45	9	81.82
Students can describe their role playing in a team and deliverables or works from their role	4	36.36	8	72.73
Students recognised their performance effects on the team accomplishment	5	45.45	9	81.82
Students delivered project outcome to solve the real world problem	4	36.36	10	90.91

Figure 1: Research Results.

Based on the research outcomes, it was confirmed that the PBL based instructional strategy was effective. It improved students analytical, problem-solving, and application skills, recognising the impact of their performance on the team's success, and achieving their project results.[4]

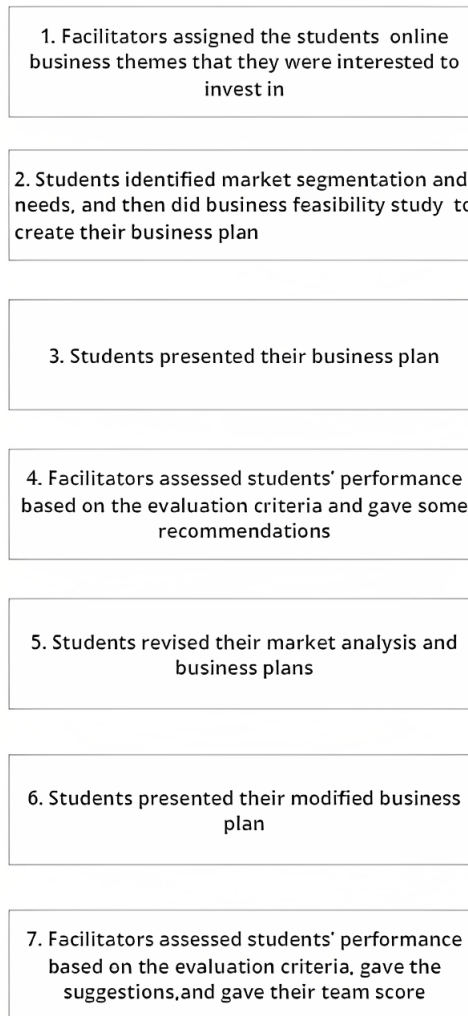


Figure 2: Research method.

2.3 PBL for Professional Training

The research conducted at the University of Minho, Portugal, focusing on students' preparedness for professional practice in PBL editions from 2005/2006 to 2007/2008. The research involved various data collection methods, including questionnaires, focus groups, interviews with students and teachers, and a document analysis.[5]

Students mentioned the positive outcome on their individual questionnaires and in the focus group discussions, as shown in the quotes below:

“Sometimes we were in classes thinking -how will this subject be useful for me in fu-

ture life –and we could actually understand the meaning and application of things.” (S.Raquel Gonçalves Fernandes,2014,p.5)

“Because there were different courses of the semester involved in the project, it made us develop the technical skills needed for the project. At the same time, we were also more interested in the different subject areas since we could see the direct application of the concepts that we were learning in the project” (S.Raquel Gonçalves Fernandes,2014,p.5)

According to teachers, PBL is having a very favourable effect at the graduate level, which is supported by the following claims:

“I think today things are increasingly influenced by the needs of the labor market, there is no doubt. And this way of learning (PBL) requires much more from an Engineer. It requires creativity! Today nothing is predictable and there are no recipes to solve problems, so either someone is sufficiently creative to find new solutions and be autonomous in their decisions... or they will not fit in the labor market. I remember when I did my Ph.D., and I interviewed several professionals from the industry and their main complaint about graduates was that they were excellent at executing things but very poor at finding solutions to new problems. They performed great academically but, in the industry, they were not capable of relating problem A, B or C to this, this and that. They were not able to take risks!” (S.Raquel Gonçalves Fernandes,2014,p.5)[5]

Artificial intelligence (AI) tools are already capable of solving straightforward, routine problems. Anyone can use AI to produce the required results by simply entering particular query questions. The fact that AI technologies can now offer solutions like code snippets and company management techniques, among other useful information, this justifies that just having basic skills won't be sufficient to secure a white-collar job in recent times. This realization makes it all the more important for graduates to have niche skills that can set them apart from such AI tools. Graduates need to acquire improved skill sets that will advance their capacity for critical thought and problem-solving. By doing this, they can increase their employability in a job market where AI is becoming more and more prevalent by doing this.

2.4 Conclusion

Project-based learning has emerged as a powerful strategy for encouraging the best learning outcomes for students and boosting their confidence to tackle challenges in a professional setup. Effective implementation of PBL requires careful planning and support. The case study [1] highlights the limitations of the peer review process that was done in order to weigh the benefits of PBL in software engineering course that includes modules like Information Systems Development (ISD), Data Structures (DS) and Web Languages and Technologies (WLT). In the case study [6] it was observed that some students gave negative feedback since there is a gap in communication between the development team and the external stakeholder expectations, it was also observed that students need to be self driven to a great extent for the PBL based learning approach.

Most of the case studies discussed above were survey-based or had a simulated project set up to enhance the learning process and make students better team players. One

of the most important incentives for students to opt for the Project Based Learning approach would be hands on projects from organizations. This can improve their resume for securing their dream job or give them the confidence to start their entrepreneurial journey

In order to motivate student to take up new challenges and switch to project based learning, an application that can bring together professors students and external stakeholders will be an ultimate learning tool. To have an unbiased assessment system both professor and external stakeholders need to collaborate and provide grades based on specific improvement criteria, to achieve this the web application will focus on providing assessment scores with justifications and suggestions for improvements. The collaboration with external stakeholders is identified as a important contributing factor to the positive impact of the course While there are challenges and limitations to PBL, these can be addressed through ongoing professional development and collaboration between teachers, students, and other stakeholders.

3 Conceptualization

Several approaches for active learning have been discussed in the previous sections however, for STEM-based courses the learning methodology needs to align with current market trends. Nowadays employers are eager to hire young minds that have analytical skills combined with a portfolio of past projects to justify the same. In order to choose the best candidates for their organizations employers try to assess candidates based on their past projects.

Students need to have a basic overview of the development lifecycles and different stages of development in order to have a smooth transition in their new organization. A curriculum enhancement that can provide students with a better view of how professional projects are developed across multinational organizations will be apt to help them get a better view of the process.

The main motivation for developing this project was to create an application that enables students to test their current skillset as well as improve their resumes by taking up new challenges. The PBL-based application will boost student's confidence as well as incentivize them by providing an opportunity to work on freelancing projects, that can help them build their professional portfolio. As students will be working on professional projects under the guidance of university professors and external stakeholders (external stakeholders will now be referred to as mentors), it provides them with an opportunity to learn from the best of both worlds

3.1 Deliverable

An application that can help students to be industry ready and give them a clear and concise picture of the basic problem-solving process to improve overall graduate employability. Based on internal project assessment scores students can get a chance to work on professional freelancing projects offered by external stakeholders that can strengthen their resume.

3.2 Platform

Considering the standardized format for study tools created by universities across the globe, the PBL-based application is a web application. The application focuses on the assignment of projects and their assessment, there will be no mobile device support as the viewport size (Screen size) is limited which can be inconvenient for the users.

3.3 Tools and Technologies

Once the platform for the web application has been finalized the next step is to decide on the tools for development. A traditional approach to development would generally include developing the user Interface in pure javascript using ES6 features with HTML, and CSS and for the backend it will be Java or C++. Although the traditional approach to development will work just fine but there are certain drawbacks worth considering before selecting the same.

For frontend development its important to keep in mind that all the webpages in the application will have multiple elements, user interaction with these element will result in DOM(Document Object Model) manipulation.The website code repository will end up having multiple JS, HTML, and CSS files.Management of all these files will be quite complex without a set of libraries or frameworks like React Js, Next.js, Angular. Instead of pure JavaScript, use of JavaScript libraries and frameworks not only makes the repository design better,it also enhances the user interactivity and provides smooth transitions

For Backend development if the traditional programming languages are used although it will provide more flexibility but the performance of the application will be hampered. This can be further justified by reports from Hackernoon [16] that suggest why popular organizations like PayPal and Netflix have migrated to Node.js from Java. It should also be noted that code complexity and productivity can have an impact based on the language and platform selected for backend development activities

Based on the above analysis the following frontend, backend, and database development, frameworks and libraries have been chosen to ensure better performance, interactivity and scalability of the application

3.3.1 Frontend

3.3.1.1 React.js

For user interface design and development, we have a range of options like Angular, Vue.js, Next.js, and React.js. I am going to choose React.js, as it's remarkably lightweight since it's a JavaScript library instead of a framework.Usually, frameworks offer limited flexibility, also community support for React is quite amazing,as its maintained by Facebook. It emphasizes the use of component-based architecture that's easy to develop and maintain. It offers a virtual DOM to enable faster rendering, which enhances the performance and user experience.[10].React also comes with JSX support that allows developers to integrate the markup and logic in a single file. The styling can be handled by styled-components. The styled-components eliminate the need to have separate CSS files for all DOM elements. React-router dom is used for seamless navigation across multiple pages throughout the application.

3.3.1.2 Material UI

Material UI is an open-source library for React components, internally it makes use of Google's material design. It provides react components like loaders cards and more, that can enhance the user-interface design. It can be easily customizable and integrated throughout the application as a plug-and-play component. The data grid from Material UI is used for grid creation with search, sort, filter, pagination, row edit, and row selection functionality.

3.3.1.3 Redux Toolkit

Generally configuring the redux store requires too much boilerplate code and additional packages are needed to use the redux store. Redux toolkit makes the redux code better by providing simplified configuration options and default middleware like `redux-thunk`. It uses Immer which allows us to write immutable updates to state in an easier way

3.3.1.4 Npm Packages

Formik is an npm package that provides form control validation and change handling. It has reusable methods, event handlers and keeps track of form state. Another popular npm package called Nivo charts is extensively used to create Pie charts and Bar charts for graphical representation of student performance

3.3.2 Backend

We have multiple frameworks/platforms for the backend to choose from like Express.js, Firebase, Prisma, Gatsby, Django, and others. Express.js is a Node.js framework designed to build APIs for web applications quickly.

3.3.2.1 Node.js

Node.js is an open-source cross-platform runtime environment for JavaScript that enables us to write JavaScript code and execute them without the need for a browser's v8 engine (used by google chrome). It has an event-driven architecture that improves the application speed. Although Java provides garbage collection at times it can take up a lot of memory. As it provides the runtime environment the backend code can be written in JavaScript, my previous experience with JavaScript will help as the learning curve will be shorter, and it can be easily linked with MySQL and MongoDB.

3.3.2.2 Express.js

It's a modular lightweight framework for building web applications. The support for middleware to execute HTTP request is one of its key features. It also provides the ability to create custom middleware. It has a built-in support for MongoDB and MySQL. It provides routing systems to handle HTTP request from different URLs

3.3.3 Database

A document database provides greater schema flexibility and reduces the need for complicated relational structure, that's why I opted for MongoDB instead of MySQL as the db does not need to be tightly coupled.

3.3.3.1 MongoDB

MongoDB provides better write performance as compared to MySQL also the schema less design makes it easy to implement and enhance in future. It also offers MongoDB Atlas that's a cloud database service that can be used free of cost. It also supports

sharding(horizontal scaling).It is an open-source database that can be easily integrated with other open-source tools and has a large and active community of developers

3.3.3.2 Mongoose

Mongoose is a nodejs module that provides functions to manipulate the documents present in MongoDB database collection. Returning updated documents or query results is simplified by Mongoose.It also includes build-in schema validation.Mongoose makes it easier to define and organize custom methods

4 Application Design Overview

It will have 3 subsection

4.1 Feature and Functionality

All features with the reason to have it

4.2 Software Architecture

The Entity Diagram and MongoDB collection

5 Implementation

5.0.1 Application flow

screenshots of UI screen and the entire application flow

5.0.2 Unit testing

6 Conclusion

7 Future Scope

...

References

- [1] P. Morais, M. J. Ferreira, and B. Veloso, “Improving student engagement with project-based learning: A case study in software engineering,” *IEEE Revista Iberoamericana de Tecnologías del Aprendizaje*, vol. 16, no. 1, pp. 21–28, 2021.
- [2] R. M. Capraro, S. W. Slough, M. M. Capraro, and J. Morgan, *STEM Project-Based Learning : An Integrated Science, Technology, Engineering, and Mathematics (STEM) Approach. Second Edition*. Boston, UNITED STATES: BRILL, 2013.
- [3] X. Weng, T. K. Chiu, and C. C. Tsang, “Promoting student creativity and entrepreneurship through real-world problem-based maker education,” *Thinking Skills and Creativity*, vol. 45, p. 101046, 2022.
- [4] N. Kurubanjerdjit and S. Nupap, “Promoting students problem solving and teamwork skills in thai undergraduates by applying direct problem-based learning strategy,” in *2021 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunication Engineering*, pp. 346–349, 2021.
- [5] S. R. G. Fernandes, “Preparing graduates for professional practice: Findings from a case study of project-based learning (pbl),” *Procedia - Social and Behavioral Sciences*, vol. 139, pp. 219–226, 2014. Educación Superior: El reto de la empleabilidad (XIII Congreso Internacional de Teoría de la Educación) Higher Education: The challenge of employability (XIII International Congress of Educational Theory).
- [6] D. Ståhl, K. Sandahl, and L. Buffoni, “An eco-system approach to project-based learning in software engineering education,” *IEEE Transactions on Education*, vol. 65, no. 4, pp. 514–523, 2022.
- [7] A. Saad and S. Zainudin, “A review of project-based learning (pbl) and computational thinking (ct) in teaching and learning,” *Learning and Motivation*, vol. 78, p. 101802, 2022.
- [8] P. Guo, N. Saab, L. S. Post, and W. Admiraal, “A review of project-based learning in higher education: Student outcomes and measures,” *International Journal of Educational Research*, vol. 102, p. 101586, 2020.
- [9] B. Pérez and A. L. Rubio, “A project-based learning approach for enhancing learning skills and motivation in software engineering,” in *Proceedings of the 51st ACM Technical Symposium on Computer Science Education, SIGCSE '20*, (New York, NY, USA), p. 309–315, Association for Computing Machinery, 2020.

The content of “Appendix” is in “\contents\app_1.tex”

A Appendix

Supplementary materials (such as source code, user menu, etc) could be included. Each appendix must be labelled (for example, Appendix A, Appendix A.1, Appendix A.2, Appendix B, Appendix B.1, etc.) and with heading. All Appendices must be referred in the text.

A.1 Points to Note

Please note the following points when you write your report:

- Consider the outline of the report. It is a good idea to start with the table of contents, which gives you an overall structure of the report.
- Show understanding of the topic and demonstrate the contribution of the work. 70% of the content of the report should be your own contributions and achievements.
- Always use your own words.
- The main report and any appendices must constitute one document.
- Pages must be numbered consecutively.
- Captions must be provided for all figures and tables.
- Equations (or important equations), figures and tables must be numbered.
- All figures and tables must be referred to in the text.
- Units of all variables must be provided.
- Numerical values (floating-point number) should be in 4 decimal places.
- Contractions should not be used.
- Check the punctuation of sentences. In particular, those sentences with equation. For example, if an equation is at the end of a sentence, a full stop should be used.
- All variables must be defined.
- Font face of variables throughout the report (in the text, equation, figures and table) must be consistent.
- Use proper headings for chapters, sections, subsections.

- Chapters, sections, subsections should be numbered and with the same numbering system throughout the report. It is suggested that vector and matrix variables should be in bold, scalar variables should be in italic.
- References must be used for materials used in the report that are not yours.
- A standard reference format must be adopted and be consistently applied through the report. General guidelines for reference format can be found on KEATS.
- Always backup your files.

B Review of stochastic calculus

B.1 Riemann integration

B.2 The Itô integral